of seven children and money was always tight. One of her tendencies is to spread her money around many different banks, and she speaks regularly about various "buckets" of money—such as one for generating her income, one for gifts to her children and grandchildren, and one for paying her bills. Spencer has been challenged by the fact that Mrs. A is quite stubborn in her opinions and rarely, if ever, listens to Spencer when he recommends that she change her way of thinking about her investment money and portfolio allocation. Her knowledge of financial concepts is limited, but she is willing to meet regularly and discuss issues with Spencer over tea.

Spencer is concerned that she is too conservative in her approach and will not accomplish one of her key goals—keeping her purchasing power—because she invests only in government bonds and cash. By taking this approach, her portfolio will not keep up with her spending after inflation and taxes in the long run; therefore, she is putting herself at risk to outlive her assets. As Spencer reflects one day on his relationship with Mrs. A, he realizes that the only recommendation she has accepted is to buy sovereign bonds to slightly increase her returns. Spencer suspects that behavioral biases are influencing Mrs. A and not permitting her to feel comfortable with changing her portfolio. Spencer asks her if she will take a 20-question assessment to examine her investor behavior. She agrees. Based on the answers to the assessment, Spencer decides to delve further into three biases: *anchoring*, *mental accounting*, and *loss aversion*. Spencer provides Mrs. A with additional questions on these three biases. Table 25.3 shows Mrs. A's answers in **bold.**

Spencer's suspicions are confirmed. Mrs. A is subject to the following biases:

- *Loss aversion bias* (the tendency to feel the pain of losses more acutely than the pleasure of gains).
- *Anchoring and adjustment bias* (the tendency to believe that current market levels are "right"; up or down directional estimates are made from the current level).
- *Mental accounting bias* (the tendency to segregate money into different "accounts."

As part of the original asset allocation process, Spencer also administered a risk tolerance questionnaire to Mrs. A for the purpose of generating a mean-variance optimization portfolio recommendation. When Spencer generated the optimization recommendation, Mrs. Alexander's "rational" asset allocation was 75 percent bonds, 15 percent stocks, and 10 percent cash; her actual allocation is 100 percent bonds. Spencer is convinced that Mrs. A needs to have a riskier portfolio than the one she currently has and that the